

Tutorial of the interactive tool to construct the restoring force triangle

The script of this interactive tool is deposited in github: <https://github.com/YunyanSun/Restoring-Force-Triangle/tree/main>.

These python codes can be directly used in Jupyter Notebook.

Introduction

Mechanophores are molecules designed to undergo specific chemical transformations in response to mechanical force. In systems where the rate-determining step involves the scission of a bond characterized by a dissociation energy ΔE and an effective bond stiffness k_{eff} , understanding how applied force influences the reaction is crucial. This analysis explores the applied force (f_a) required to facilitate the reaction when the thermal energy barrier is fixed to user-specified values of activation energy ($\Delta E^\ddagger$).

By modeling the system using the Restoring Force Triangle (RFT) and Reaction Coordinate Diagrams (RCD), we can predict the necessary applied force to achieve a desired activation energy. The RFT provides a graphical representation of the mechanical work performed on the bond as it stretches under force until reaching the transition state. The RCD illustrates the changes in energy along the reaction pathway, showing how mechanical force alters the energy landscape.

Understanding these relationships has implications for the design of mechanophores and control of mechanochemical reactions. It allows for the prediction and manipulation of reaction rates under mechanical stress, which is valuable in fields such as materials science, polymer chemistry, and biomechanics.

Disclaimer: *It is important to note that the RFT is an approximation and is, at best, semi-quantitative.* The primary value of the RFT lies in building intuition about force-facilitated reactions rather than providing precise quantitative predictions. This tool is intended to aid in understanding the general trends and relationships between applied force, activation energy, and reaction pathways in mechanochemical processes.

User Inputs

The analysis requires the following inputs:

1. ΔE (**kcal/mol**): The bond dissociation energy, representing the intrinsic energy barrier for bond scission without applied force. It defines the bond's inherent stability in the absence of mechanical intervention.
2. k_{eff} (**nN/Å**): The effective spring constant, describing the bond's stiffness and resistance to deformation under force. This molecular descriptor plays a key role in understanding mechanochemical coupling by quantifying how efficiently tensile force is transmitted to the bond being stretched.
3. $\Delta E^\ddagger$ values (**kcal/mol**): Up to three activation energy values under applied force, allowing for the exploration of force-facilitated reactions at different energy barriers. **The default value is 16.0 kcal/mol**, which corresponds to the typical thermal energy barrier surmounted in single-molecule force spectroscopy experiments.

Note: The values for ΔE and k_{eff} are computed using the methods described in the supplemental information of the following publication: [Yunyan Sun et al., Chem, 2024, DOI: 10.1016/j.chempr.2024.05.012](https://doi.org/10.1016/j.chempr.2024.05.012).

1. Input for ΔE

The bond dissociation energy, ΔE , is entered in kilocalories per mole (kcal/mol).

- **Parameter:** ΔE
- **Units:** kcal/mol
- **Description:** Energy to bring about bond scission without applied force.

2. Input for k_{eff}

The effective spring constant k_{eff} represents the stiffness of the bond being broken. This value is entered in nanonewtons per angstrom (nN/Å).

- **Parameter:** k_{eff}
- **Units:** nN/Å
- **Description:** Effective stiffness of the bond.

3. Input for $\Delta E^\ddagger$ Values

The activation energy under force, $\Delta E^\ddagger$, is the user-specified thermal energy barrier when the bond is under an applied force. The default value is set to **16.0 kcal/mol**, but the user can choose to provide up to three custom values.

- **Parameter:** $\Delta E^\ddagger$
- **Units:** kcal/mol
- **Description:** User-specified activation energy under applied force.
- **Options:**
 - **Default:** Accept the default value of 16.0 kcal/mol.
 - **Single Custom Value:** Provide one custom value for $\Delta E^\ddagger$.
 - **Comparison Mode:** Provide two or three custom values to compare how different $\Delta E^\ddagger$ values affect the required applied force and other parameters.

Selection in Comparison Mode

If multiple $\Delta E^\ddagger$ values are provided, the program enters **comparison mode**. In this mode:

- The plot of $\Delta E^\ddagger$ vs. applied force f_a will display horizontal lines at each specified $\Delta E^\ddagger$ value.
- Each $\Delta E^\ddagger$ value will be represented by a unique color for clear distinction.
- Points of intersection between these lines and the $\Delta E^\ddagger$ vs. f_a curve will indicate the required applied force f_a for each activation energy.

Restoring Force Triangle (RFT) Plot:

- Only one RFT will be plotted, corresponding to a selected $\Delta E^\ddagger$ value.
- The user will choose which $\Delta E^\ddagger$ value should be used for the RFT plot.
- The chosen $\Delta E^\ddagger$ will determine the applied force f_a and other parameters used to construct the RFT.

Understanding Comparison Mode:

- By comparing different $\Delta E^\ddagger$ values, the non-linear dependence of f^* on $\Delta E^\ddagger$ becomes evident.
- This analysis also highlights how the RFT subregions, as well as the positions of X_{TAB} and X_{TTS} , change with varying $\Delta E^\ddagger$ values.

Proceeding with the Analysis

After providing the necessary inputs, the program will:

- Calculate the maximum force f_{max} (the applied force which reduces $\Delta E^\ddagger$ to zero (i.e., barrierless)).
- Compute the required applied force f_a for each $\Delta E^\ddagger$ value using the relationship:

$$f_a = f_{max} \left(1 - \sqrt{\frac{\Delta E^\ddagger}{\Delta E}}\right)$$

- Generate plots:
 - **$\Delta E^\ddagger$ vs. Applied Force Plot:** Visualizes how the activation energy decreases with increasing applied force.
 - **Restoring Force Triangle (RFT) Plot:** Illustrates the mechanical work done on the bond at the selected $\Delta E^\ddagger$ value.
 - **Reaction Coordinate Diagrams (RCDs):** Compares the reaction pathways under different applied forces within the limits of the RFT.
- Create tables of parameters and energies, including:
 - Common parameters shared across all $\Delta E^\ddagger$ values.
 - Parameters and energies specific to each $\Delta E^\ddagger$ value.
- Perform validation checks to ensure the calculations are consistent and accurate.

Calculations

In this section, we outline the calculations performed to determine the required applied force f^* for each specified activation energy $\Delta E^\ddagger$, as well as other key parameters related to the Restoring Force Triangle (RFT) and the Reaction Coordinate Diagrams (RCDs).

1. Conversion of Energy Units

The bond dissociation energy ΔE is initially provided in kilocalories per mole (kcal/mol). For consistency in calculations involving force and displacement, we convert ΔE to nanonewton·angstroms (nN·Å) using the conversion factor:

- **Conversion Factor:** 1 kcal/mol=0.069477nN·Å

Calculation:

$$E = \Delta E \times \text{Conversion Factor}$$

Where:

- E is the energy in nN·Å.
- ΔE is the bond dissociation energy in kcal/mol.

2. Calculation of Maximum Force $f_{\max}$

The maximum force that can be applied to a bond, reducing the activation energy $\Delta E^\ddagger$ to zero, such that bond rupture becomes barrierless. At this force, no thermal energy is required to break the bond, meaning it fails purely due to the applied mechanical force.

Formula:

$$f_{\max} = \sqrt{\frac{2 \cdot k_{\text{eff}} \cdot E}{9}}$$

Where:

- $f_{\max}$ is the force that brings about bond scission spontaneously (i.e., barrierless).
- k_{eff} is the effective spring constant of the bond (in nN/Å), representing the bond's stiffness.
- E is the bond dissociation energy (in nN·Å), which is related to the energy required to break the bond in the absence of force.

Explanation:

$f_{\max}$ is the altitude of the RFT; the formula derives from the geometric features of the RFT.

3. Calculation of Predicted Applied Force f^* for Each $\Delta E^\ddagger$

For each specified activation energy $\Delta E^\ddagger$, we calculate the required applied force f^* to reduce the energy barrier to $\Delta E^\ddagger$.

Formula:

$$f_a = f_{max} \left(1 - \sqrt{\frac{\Delta E^\ddagger}{\Delta E}}\right)$$

Where:

- f_a is the predicted applied force in nN.
- $\Delta E^\ddagger$ is the desired activation energy under applied force in kcal/mol.
- ΔE is the total energy barrier without applied force in kcal/mol.

Explanation:

- The term $\sqrt{\frac{\Delta E^\ddagger}{\Delta E}}$ represents the fraction of the energy barrier remaining under the applied force.
- Subtracting this fraction from 1 gives the proportion of the maximum force needed to achieve the desired reduction in the energy barrier.
- Multiplying by f_{max} provides the required applied force f_a .

4. Calculation of Thermal Energy Barrier $\Delta E^\ddagger$ vs. Applied Force f_a

The relationship between the thermal energy barrier $\Delta E^\ddagger$ and the applied force f_a is given by:

Formula:

$$\Delta E^\ddagger = \Delta E * \left(1 - \frac{f_a}{f_{max}}\right)^2$$

Where:

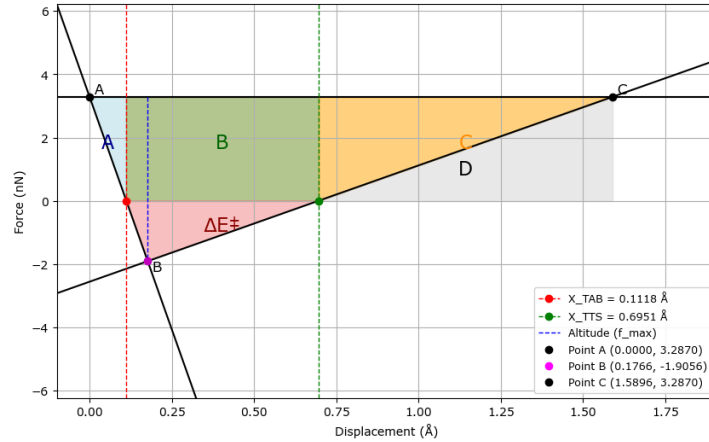
- $\Delta E^\ddagger$ is the thermal energy barrier under applied force.
- f_a is the applied force (varies from 0 to f_{max}).
- ΔE and f_{max} are as previously defined.

Explanation:

- This quadratic relationship shows how the thermal energy barrier decreases with increasing applied force.
- The term $(1 - f_a/f_{max})$ represents the normalized force, and squaring it reflects the energy dependence.

5. Defining the Restoring Force Triangle (RFT)

The Restoring Force Triangle (RFT) is a geometric construct that represents the balance between the restoring force a stretched bond and the applied tensile force. The area of the RFT corresponds to the bond dissociation energy ΔE .



The RFT is defined by three intersecting lines:

I. **Line 1 (Base of the RFT):**

$$Y = f_a$$

- This horizontal line represents the applied force f_a and forms the base of the triangle.

II. **Line 2:**

$$Y = -k_{\text{eff}} * X + f_a$$

- This line has a negative slope, reflecting the effective stiffness of the scissile bond, k_{eff} . It intersects Line 1 at point A and Line 3 at point B.

III. **Line 3:**

$$Y = k_{\text{eff}}/8 * X + d + f_d$$

- Line 3 has a positive slope and represents the "let-off" restoring force component. It intersects Line 1 at point C and Line 2 at point B. The intercept d is calculated based on k_{eff} and ΔE .

Using the calculated applied force f^* , we determine the coordinates of the key points (A, B, and C) that define the RFT.

Point A:

- **Coordinates:** (X_A, Y_A)
- **Values:**
 - $X_A = 0$
 - $Y_A = f_a$

Point B:

- **Coordinates:** (X_B, Y_B)
- **Values:**
 - $X_B = -8*d/9 * k_{eff}$
 - $Y_B = -k_{eff} * X_B + f_a$

Point C:

- **Coordinates:** (X_C, Y_C)
- **Values:**
 - $X_C = -8*d*k_{eff}$
 - $Y_C = f_a$

Calculation of Intercept d :

The intercept d is calculated using: $d = -\sqrt{\frac{9k_{eff}*E}{32}}$

Explanation:

- The intercept c is derived from the geometric properties of the RFT and relates to the curvature of the potential energy surface.
- It is essential for determining the positions of Points B and C.

6. Calculation of Intersection Points X_{TAB} and X_{TTS}

X_{TAB} :

- Represents the displacement at which the applied force line intersects the x-axis (point where $Y=0$).
- **Formula:**

$$X_{TAB} = \frac{f_a}{k_{eff}}$$

X_{TTS} :

- Represents the displacement corresponding to the Tensioned Transition State (TTS).
- **Formula:**

$$X_{TTS} = -\frac{8*(d+f_a)}{k_{eff}}$$

7. Calculation of Subregion Energies in the RFT

The RFT is divided into several subregions, each representing mechanical work or energy contributions. We calculate the energies associated with each subregion.

Energy of Triangle A:

- **Formula:**

$$E_{Triangle A} = \frac{1}{2} * X_{TAB} * \frac{f_a}{conversion factor}$$

- **Explanation:**

- Represents the mechanical work done on the bond in the GS to reach the Tension-Activated Bond (TAB).

Energy of Rectangle B:

- **Formula:**

$$E_{Rectangle B} = (X_{TTS} - X_{TAB}) * \frac{f_a}{conversion factor}$$

- **Explanation:**

- Represents the mechanical work done on the bond between the TAB and the TTS at constant force.

Energy of Triangle $\Delta E^\ddagger$:

- **Formula:**

$$\Delta E^\ddagger = \Delta E * (1 - \frac{f_a}{f_{max}})^2$$

- **Explanation:**

- Calculated using the shoelace formula or geometric methods.
- Represents the activation energy under applied force.

Energy of Triangle C:

- **Formula:**

$$E_{Rectangle C} = \frac{1}{2} (X_c - X_{TTS}) * \frac{f_a}{conversion factor}$$

- **Explanation:**

- This energy represents the mechanical work performed on the bond after the transition state (TTS) as the bond continues to stretch under the applied force. It reflects the energy internal to the scissile bond during this phase of the reaction.

Energy of Triangle D:

- **Formula:**

$$E_{Rectangle D} = \frac{1}{2} (X_c - X_{TTS}) * \frac{f_a}{conversion factor}$$

- **Explanation:**

- The energy of Triangle D is assigned a **negative value**, representing the **surplus mechanical energy** acquired by the molecule as it moves from the Tensioned Transition State (TTS) to the Tensioned Intermediate (TI) under the constant applied force f^* .
- **Mechanism of Energy Transfer:**
 - After the TTS, the applied force continues to act on the molecule, performing work over the displacement from X_{TTS} to X_C .
 - This surplus mechanical work is converted into **kinetic energy** of the molecular constituents.
- **Influence on Reaction Dynamics:**
 - The surplus kinetic energy imparts **momentum to the atoms** surrounding the scissile bond.
 - This can channel the reaction trajectory into a '**flyby**' **mode**, where the reaction products possess sufficient kinetic energy to bypass intermediate states or avoid intermediate recombination.
 - Such **nonstatistical dynamics** can influence the **product distribution**, leading to outcomes that deviate from predictions based on conventional transition state theory.
 - This phenomenon is discussed in the study by **Yun Liu et al.:**

Reference:

Yun Liu et al., "Flyby reaction trajectories: Chemical dynamics under extrinsic force." *Science*, 373(6552), 208–212 (2021). <https://doi.org/10.1126/science.abi7609>

- **Significance in Transition State Theory:**
 - In the context of transition state theory, the additional kinetic energy from the surplus mechanical work can lead to deviations from the expected distribution of reaction products.
 - The reaction pathway may be altered due to this kinetic energy, resulting in nonstatistical behavior that affects product formation and selectivity.
 - Understanding these effects is crucial for accurately modeling mechanochemical reactions and predicting their outcomes.

8. Validation Checks

Total Area of the RFT:

- The total area enclosed by the RFT should correspond to the original energy barrier ΔE .
- **Verification:**

$$\text{Total Area} = E_{\text{Triangle A}} + E_{\text{Rectangle B}} + E_{\Delta E^\ddagger} + E_{\text{Triangle C}}$$

- **Check:**

$$\text{Total Area} \approx \Delta E$$

Area Corresponding to $\Delta E^\ddagger$:

- The area of the sub-triangle representing $\Delta E^\ddagger$ should correspond to the specified activation energy under applied force.
- **Verification:**

$$E_{\Delta E^\ddagger} \approx \Delta E^\ddagger$$

9. Calculation of Energies for Reaction Coordinate Diagrams (RCDs)

Using the subregion energies, we calculate the energy levels for different states in the RCD.

Ground State (GS):

- **Energy:**

$$E_{\text{GS}} = 0 \text{ kcal/mol}$$

Tension-Activated Bond (TAB):

- **Energy:**

$$E_{\text{TAB}} = E_{\text{Triangle A}}$$

Tensioned Transition State (TTS):

- **Energy:**

$$E_{\text{TTS}} = E_{\text{TAB}} + \Delta E^\ddagger$$

Tensioned Intermediate (TI):

- **Energy:**

$$E_{\text{TI}} = E_{\text{TTS}} + E_{\text{Triangle C}}$$

10. Comparison of Multiple $\Delta E^\ddagger$ Values (If Applicable)

When multiple $\Delta E^\ddagger$ values are provided, the calculations are repeated for each value. This allows for the comparison of how different activation energies under applied force affect:

- The required applied force f^* .
- The positions of Points A, B, C.
- The energies of the subregions in the RFT.
- The energy levels in the RCDs.

Note on Units:

- Energies calculated in $\text{nN} \cdot \text{\AA}$ are converted back to kcal/mol using the conversion factor for consistency.
- Displacements are measured in angstroms ($\text{\AA}$).
- Forces are measured in nanonewtons (nN).

Plots

In this section, we present and explain the plots generated from the calculations. These visual representations aid in understanding the relationships between the applied force, activation energy, and the mechanical work involved in the reaction process.

1. Plot of $\Delta E^\ddagger$ vs. Applied Force f_a

The plot of $\Delta E^\ddagger$ vs. applied force f_a illustrates how the activation energy decreases as the applied force increases. This relationship is crucial in mechanochemistry, where applying force can facilitate reactions by lowering the energy barrier.

Description:

- **X-axis:** Applied force f_a (in nN).
- **Y-axis:** Activation energy $\Delta E^\ddagger$ (in kcal/mol).
- **Curve:** Represents the quadratic relationship between $\Delta E^\ddagger$ and f_a , given by:

$$\Delta E^\ddagger = \Delta E * (1 - \frac{f_a}{f_{max}})^2$$

- **Horizontal Lines:** If multiple $\Delta E^\ddagger$ values are specified, horizontal dashed lines are drawn at these values to indicate target activation energies.
- **Intersection Points:** The points where the horizontal lines intersect the curve represent the required applied force f^* to achieve the corresponding $\Delta E^\ddagger$.

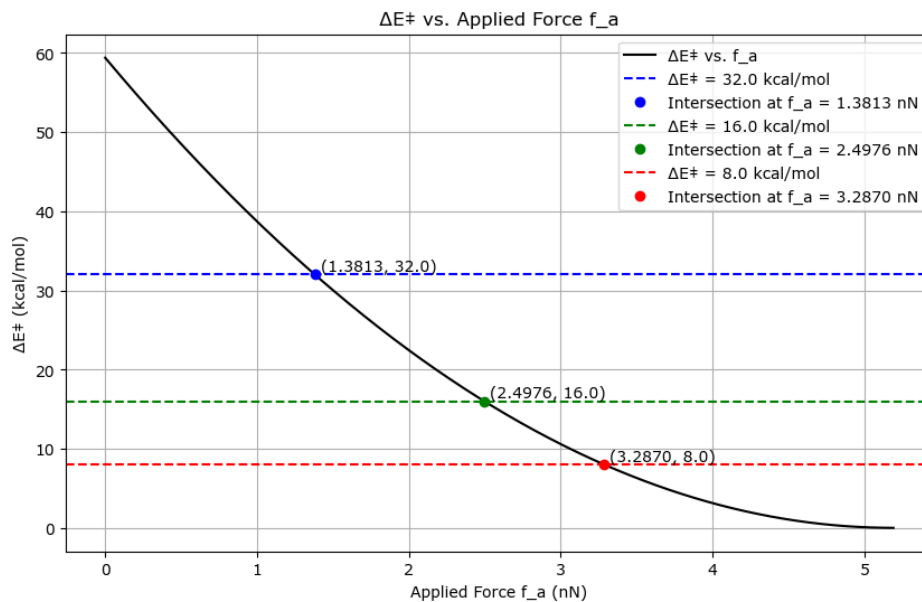
Interpretation:

- As f_a increases from 0 to f_{max} , the activation energy $\Delta E^\ddagger$ decreases from ΔE to 0.
- The plot allows us to visually determine the applied force needed to reach a desired activation energy.
- Intersection points provide specific f^* values for each $\Delta E^\ddagger$, which are crucial for further calculations.

Visualization:

- **Colors:** Each $\Delta E^\ddagger$ value is represented with a unique color for clarity.
- **Annotations:** Intersection points are marked and labeled with their coordinates.

$\Delta E^\ddagger$ vs. f_a Plot:



2. Restoring Force Triangle (RFT) Plot

- The RFT plot provides a graphical representation of the mechanical work performed on the bond as it stretches under the applied force.

Description:

- Axes:**
 - X-axis:** Displacement (in Å).
 - Y-axis:** Force (in nN).
- Lines:**
 - Line 1:** Horizontal line representing the constant applied force f^* .
 - Line 2:** Line with slope $-k_{\text{eff}}$, representing the bond's restoring force.
 - Line 3:** Line representing the transition from the TTS to the TI.
- Points:**
 - Point A:** (X_A, Y_A) — starting point at $X=0$ and $Y=f_a$.
 - Point B:** (X_B, Y_B) — intersection of Line 2 and Line 3.
 - Point C:** (X_C, Y_C) — endpoint at $X=X_C$ and $Y=f_a$.
- Subregions:** The area under the lines is divided into subregions representing different energy contributions:
 - Triangle A:** Electronic energy difference to reach the TAB.
 - Rectangle B:** Mechanical work between TAB and TTS at constant force.
 - Triangle $\Delta E^\ddagger$:** Activation energy under applied force.
 - Triangle C:** Mechanical work after the TTS as the bond continues to stretch.
 - Triangle D:** Negative energy representing the surplus of mechanical energy acquired as the system moves from the TTS to the TI.

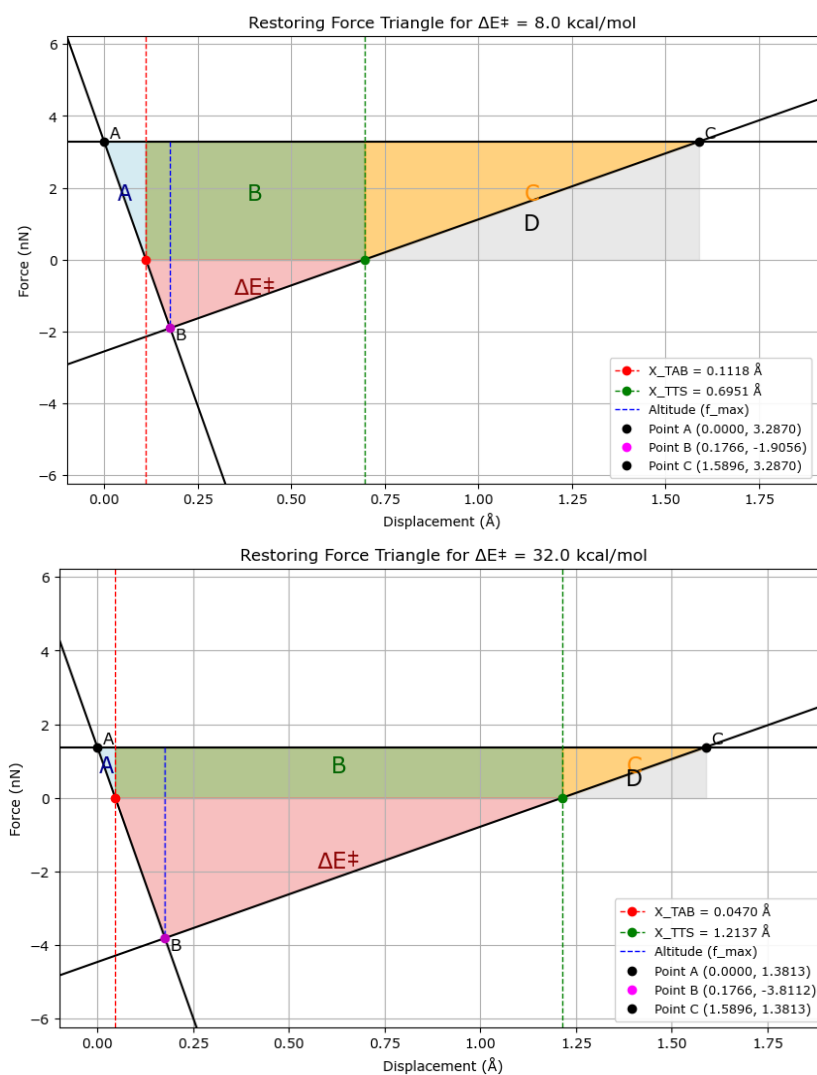
Interpretation:

- The total area under the curve corresponds to the bond dissociation energy ΔE .
- Each subregion's area represents a specific energy contribution, aiding in understanding how mechanical work facilitates the reaction.
- The RFT visually demonstrates how applying force modifies the energy landscape of the reaction.

Visualization:

- **Shading:** Different colors or shades are used to distinguish subregions.
- **Annotations:** Points A, B, C, XTAB, and XTTS are marked and labeled.
- **Legend:** A legend explains the significance of each line and subregion.

Restoring Force Triangle Plots:



3. Reaction Coordinate Diagrams (RCDs)

Reaction Coordinate Diagrams illustrate the energy changes that occur during the reaction process, showing the progression from reactants to products through various intermediates and transition states.

Description:

- **X-axis:** Reaction coordinate or displacement (in Å).
- **Y-axis:** Energy (in kcal/mol).
- **States:**
 - **Ground State (GS):** Starting point with energy $E_{GS}=0$.
 - **Tension-Activated Bond (TAB):** Energy $E_{TAB}=E_{\text{Triangle A}}$.
 - **Tensioned Transition State (TTS):** Energy $E_{TTS}=E_{TAB}+\Delta E^\ddagger$.
 - **Tensioned Intermediate (TI):** Energy $E_{TI}=E_{TTS}+E_{\text{Triangle D}}$.
- **Paths:**
 - Lines connecting the states represent the progression of the reaction under applied force.
- **Multiple $\Delta E^\ddagger$ Values:**
 - If multiple $\Delta E^\ddagger$ values are specified, their corresponding RCDs are plotted on the same graph for comparison.
 - Each pathway is color-coded to match the colors used in the $\Delta E^\ddagger$ vs. f_a plot.

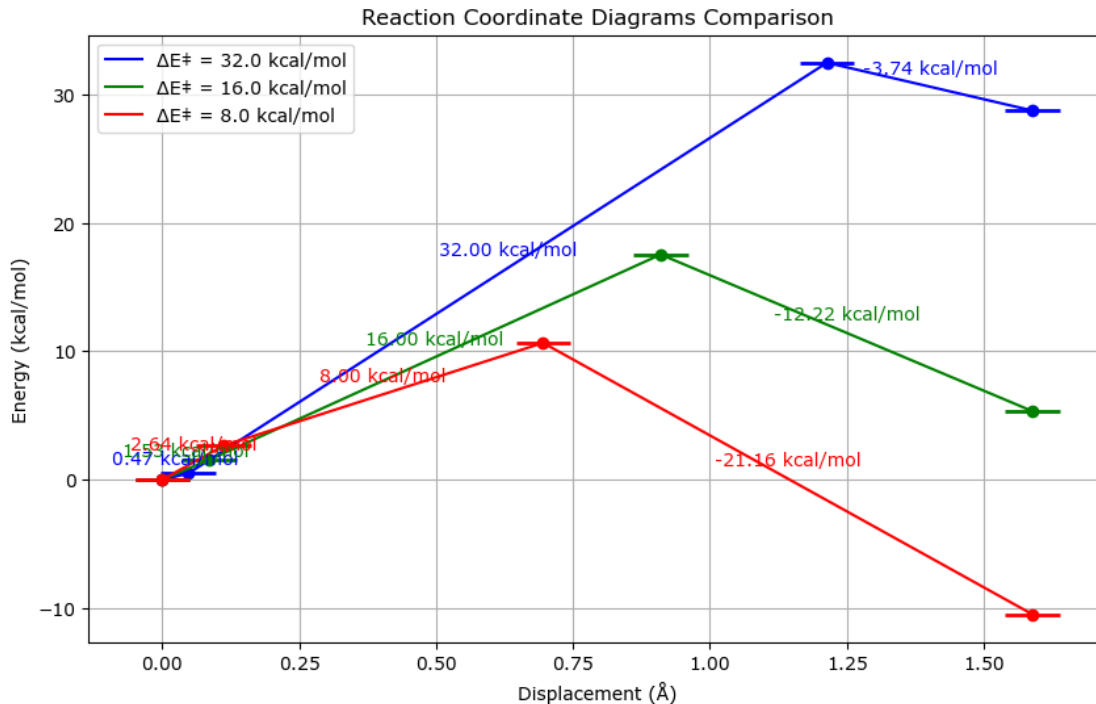
Interpretation:

- The diagrams show how the energy levels of intermediates and transition states change with different applied forces.
- Comparing RCDs for different $\Delta E^\ddagger$ values highlight the impact of force on the reaction pathway.
- The energy differences between states provide insight into the mechanical work and energy barriers involved.

Visualization:

- **Colors:** Each $\Delta E^\ddagger$ value/pathway is represented with a unique color consistent across plots.
- **Annotations:**
 - Energy levels are labeled with their corresponding values.
 - Energy differences between states are annotated.
- **Legend:** Explains the color-coding and labels for different pathways.

Reaction Coordinate Diagrams:



4. Combined Analysis

By examining all the plots together, we gain a comprehensive understanding of how mechanical force influences the reaction:

- **Applied Force vs. Activation Energy:** Demonstrates the theoretical relationship and the specific forces required for desired activation energies.
- **RFT:** Provides a geometric interpretation of the mechanical work done on the bond, correlating with the areas representing energy contributions.
- **RCDs:** Show the changes in energy landscapes for different applied forces, emphasizing the effect on reaction intermediates and transition states.

Note on Interpretation:

- The plots are interconnected; understanding one enhances the comprehension of the others.
- They collectively illustrate the semi-quantitative nature of the RFT model, emphasizing its value in building intuition about force-facilitated reactions.

Results:

In this section, we present the results obtained from the calculations based on the user-provided inputs. The results include tables of common parameters, $\Delta E^\ddagger$ -dependent parameters, and observations derived from the calculations and plots.

1. Common Parameters

Using a prototypical mechanophore, the following table summarizes the parameters that are common across all calculations:

Parameter	Symbol	Value	Units
Total Energy Barrier	ΔE	59.4	kcal/mol
Effective Spring Constant	k_{eff}	29.4	nN/Å
Maximum Force	f_{max}	5.19	nN

Explanation:

- ΔE : The bond dissociation energy.
- k_{eff} : The stiffness of the bond being broken.
- f_{max} : The maximum force that can be applied to the bond such that bond failure becomes barrierless. At this force, the activation energy $\Delta E^\ddagger$ is reduced to zero, meaning the bond can break without any thermal energy.

$$f_{\text{max}} = \sqrt{\frac{2 * k_{\text{eff}} * E}{9}}$$

where $E = \Delta E \times \text{Conversion Factor}$.

2. $\Delta E^\ddagger$ -Dependent Parameters

The table below presents the parameters that depend on each specified activation energy $\Delta E^\ddagger$.

Table of Parameters for Each $\Delta E^\ddagger$

Parameter	Units	$\Delta E^\ddagger = 32.0$ kcal/mol	$\Delta E^\ddagger = 16.0$ kcal/mol	$\Delta E^\ddagger = 8.0$ kcal/mol
Required Applied Force f^*	nN	1.38	2.50	3.29
Point A (X_A, Y_A)	Å, nN	(0, 1.38)	(0, 2.50)	(0, 3.29)
Point B (X_B, Y_B)	Å, nN	(0.18, -3.81)	(0.18, -2.70)	(0.18, -1.91)
Point C (X_C, Y_C)	Å, nN	(1.59, 1.38)	(1.59, 2.50)	(1.59, 3.29)
Displacement X_{TAB}	Å	0.047	0.085	0.11
Displacement X_{TTS}	Å	1.21	0.91	0.70
Energy of Triangle A	kcal/mol	0.47	1.53	2.65
Energy of Rectangle B	kcal/mol	23.2	29.7	27.6
Energy of Triangle $\Delta E^\ddagger$	kcal/mol	32.0	16.0	8.00
Energy of Triangle C	kcal/mol	3.74	12.2	21.2
Energy of Triangle D	kcal/mol	-3.74	-12.2	-21.2

Explanation:

- **Required Applied Force f_a** : Calculated using:

$$f_a = f_{max} (1 - \sqrt{\frac{\Delta E^\ddagger}{\Delta E}})$$

- **Points A, B, C:** Coordinates defining the Restoring Force Triangle.
- **Displacements XTAB, XTTS:** Positions corresponding to the Tension-Activated Bond (TAB) and Tensioned Transition State (TTS).
- **Energies of Subregions:** Mechanical work or energy contributions represented by different areas in the RFT.

3. Observations

Based on the results obtained, we make the following observations:

Applied Force vs. Activation Energy

- **Trend:** As the desired activation energy $\Delta E^\ddagger$ decreases, the required applied force f_a increases.
- **Interpretation:** Applying a higher force lowers the activation energy, facilitating the reaction.

Restoring Force Triangle Parameters

- **Points A, B, C:** The positions of these points shift with varying f^* , altering the shape of the RFT.
- **Subregion Energies:**
 - **Triangle A:** Increases with higher f_a , representing more mechanical work to reach TAB.
 - **Triangle $\Delta E^\ddagger$:** Corresponds closely to the specified activation energy, validating the calculations.
 - **Triangle D:** The negative energy becomes more pronounced with higher f_a , indicating more surplus kinetic energy imparted to the molecular fragments.

Reaction Coordinate Diagrams

- **Energy Levels:**
 - The energy at TTS (E_{TTS}) increases with higher f^* due to the added mechanical work.
 - The energy difference between TTS and TI ($E_{\text{Triangle D}}$) becomes more negative, reflecting the increased surplus kinetic energy.
- **Comparative Analysis:**
 - Plotting multiple RCDs shows how the reaction pathway changes under different applied forces.
 - Higher applied forces lead to lower activation energies and altered energy landscapes.

4. Validation Checks

Total Energy Consistency

- **Verification:**

$$\text{Total Energy} = E_{\text{Triangle A}} + E_{\text{Rectangle B}} + E_{\Delta E^\ddagger} + E_{\text{Triangle C}}$$

- **Result:** The total calculated energy closely matches the original energy barrier ΔE , confirming the accuracy of the calculations.

Activation Energy Correspondence

- **Verification:**

$$E_{\Delta E^\ddagger} \approx \Delta E^\ddagger$$

- **Result:** The energy of Triangle $\Delta E^\ddagger$ closely matches the specified activation energy for each case, validating the method.

5. Discussion

- **Impact of Applied Force:**

- Applying force reduces the activation energy, making the reaction more favorable.
- The required force to achieve a certain reduction in activation energy depends on the bond stiffness and the original energy barrier.

- **Mechanical Work and Kinetic Energy:**

- The mechanical work done on the bond is partitioned into different energy contributions.
- Surplus mechanical energy (Triangle D) can lead to increased kinetic energy in the reaction products, potentially affecting reaction dynamics.

- **Nonstatistical Reaction Dynamics:**

- The surplus kinetic energy may result in nonstatistical behavior, such as 'flyby' trajectories, influencing product distribution.
- This highlights the importance of considering mechanical effects in reaction modeling.

6. Conclusion of Results

The calculations and visualizations demonstrate the relationship between applied force and activation energy in mechanochemical reactions. By adjusting the applied force, it is possible to modulate the activation energy and influence the reaction pathway. The Restoring Force Triangle provides a useful framework for understanding the mechanical work involved, and the Reaction Coordinate Diagrams illustrate the changes in energy landscapes under different forces.

Parity Plot: Experimental vs Predicted f_a

To assess the accuracy of the model, we compared the predicted values of f^* predicted against experimentally determined f^* experimental values (from single molecule force spectroscopy) for a series of compounds. We constructed a **parity plot**, which is a standard way to visualize how well the predictions align with the experimental data.

$$f^* = f_{max} \left(1 - \sqrt{\frac{16.0}{\Delta E}}\right)$$

The predicted force $f^*_{\text{predicted}}$ is calculated as the force required to reduce the energy barrier to 16 kcal/mol (around the typical activation barrier in single-molecule force spectroscopy experiments), based on transition state theory.

Each point in the parity plot represents a specific compound, where the x-axis shows the experimental f^* values, and the y-axis shows the predicted f^* values. A dashed line indicates perfect agreement between the two.

Statistical Analysis of the Parity Plot

To quantify the agreement between the experimental and predicted values, we computed the following statistics:

Mean Squared Error (MSE): This measures the average squared difference between the predicted and actual values.

$$MSE = \frac{1}{n} * \sum_{i=1}^n (f^*_{\text{experiment}} - f^*_{\text{predicted}})^2$$

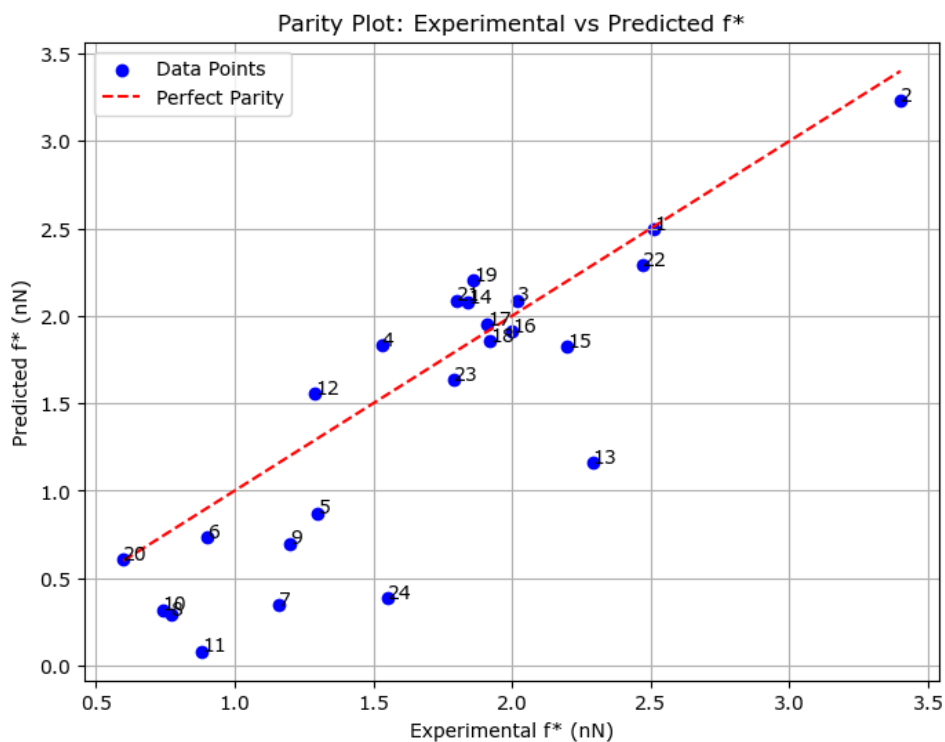
Root Mean Squared Error (RMSE): This is the square root of the MSE and gives a sense of the average magnitude of the errors in the same units as f^* .

$$RMSE = \sqrt{MSE}$$

R-squared (R^2): This statistic provides a measure of how well the predicted values explain the variance in the experimental data. An R^2 value closer to 1 indicates better predictive accuracy.

$$R^2 = 1 - \frac{\sum_{i=1}^n (f^*_{\text{experiment}} - f^*_{\text{predicted}})^2}{\sum_{i=1}^n (f^*_{\text{experiment}} - \bar{f}^*_{\text{predicted}})^2}$$

We employ mechanophores **1-24** as an example (see figure 9 in the main text).



The results of the parity plot analysis were as follows:

- **MSE:** 0.2277
- **RMSE:** 0.4772 nN
- **R-squared (R^2):** 0.4686

Despite the simplicity of the model, the agreement between the experimental and predicted values is surprisingly good. The R^2 value indicates that the model captures nearly half of the variance in the data, which is promising for such a straightforward approach.

For a more quantitative, predictive, yet less straightforward approach, please see our previous work in *Chem.* **2024**, *10*, 3055–3066 DOI: 10.1016/j.chempr.2024.05.012.